

GM-FBG-16/32 Fiber Bragg Grating Interrogator

Industrial FBG wavelength interrogation platform · C-band 1527~1564 nm

The GM-FBG series is an industrial-grade FBG wavelength interrogation platform for structural health monitoring: ± 2 pm typical wavelength accuracy, ± 1 pm repeatability and 40 dB dynamic range; wide temperature and supply range with low power draw; a compact 161×144×33 mm chassis for embedded integration and field deployment. Dozens of gratings with different center wavelengths can be daisy-chained on a single fiber, so one unit forms a quasi-distributed multi-point network.



GM-FBG-16/32 interrogator

- Bridge & tunnel SHM
- Wind blades & hybrid towers
- Power equipment temperature
- Mine roof monitoring (with flameproof processor)

Features

- ± 2 pm typical wavelength accuracy, ± 1 pm repeatability and stability
- 40 dB dynamic range with margin for long chains of gratings
- 1~100 Hz scan rate; high-speed scanning on request
- RJ45 / RS485 / CAN / 2.4G wireless for flexible field networking
- DC 9~36 V wide-range supply, ≤ 10 W total consumption
- 161×144×33 mm compact chassis, cabinet or on-site mounting
- Compatible with third-party C-band grating sensors

Specifications

Optical performance	
Optical channels	1~32 optional
Wavelength range	1527~1564 nm
Wavelength accuracy	± 2 pm (typical)
Repeatability / stability	± 1 pm / ± 1 pm
Dynamic range	40 dB
Scan rate	1~100 Hz (high speed on request)
Electrical & interfaces	
Interfaces	RJ45 / RS485 / CAN / 2.4G wireless
Power supply	DC 9~36 V
Consumption	≤ 10 W
Environment & mechanical	
Operating temperature	-20~75 °C
Storage temperature	-40~80 °C

Ordering information

Model	Configuration	Notes
GM-FBG-16	16 optical channels	Standard unit for small and mid-size arrays
GM-FBG-32	32 optical channels	Large-scale networking unit
Custom	Channels / scan rate	High-speed sampling and special channel counts per project

Network topology

GM-F grating sensors in series ×N → GM-FBG-16/32 interrogator → RJ45 / RS485 / CAN / 2.4G → SHM platform / user system;
compatible with third-party C-band sensors.

Warranty & service

2 yr Hardware warranty · spares stocked	24 h On-site response, East China	48 h On-site response, nationwide	7×24 Remote diagnostics hotline
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Standard service policy; the supply contract prevails.

Certifications & standards

 China National Accreditation Service for Conformity Assessment CNAS testing	 计量认证 CMA metrology	 ISO 9001 quality	 ISO 14001 environmental	 矿用产品安全标志 MA mining safety
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Certificate numbers and scans are available on request.